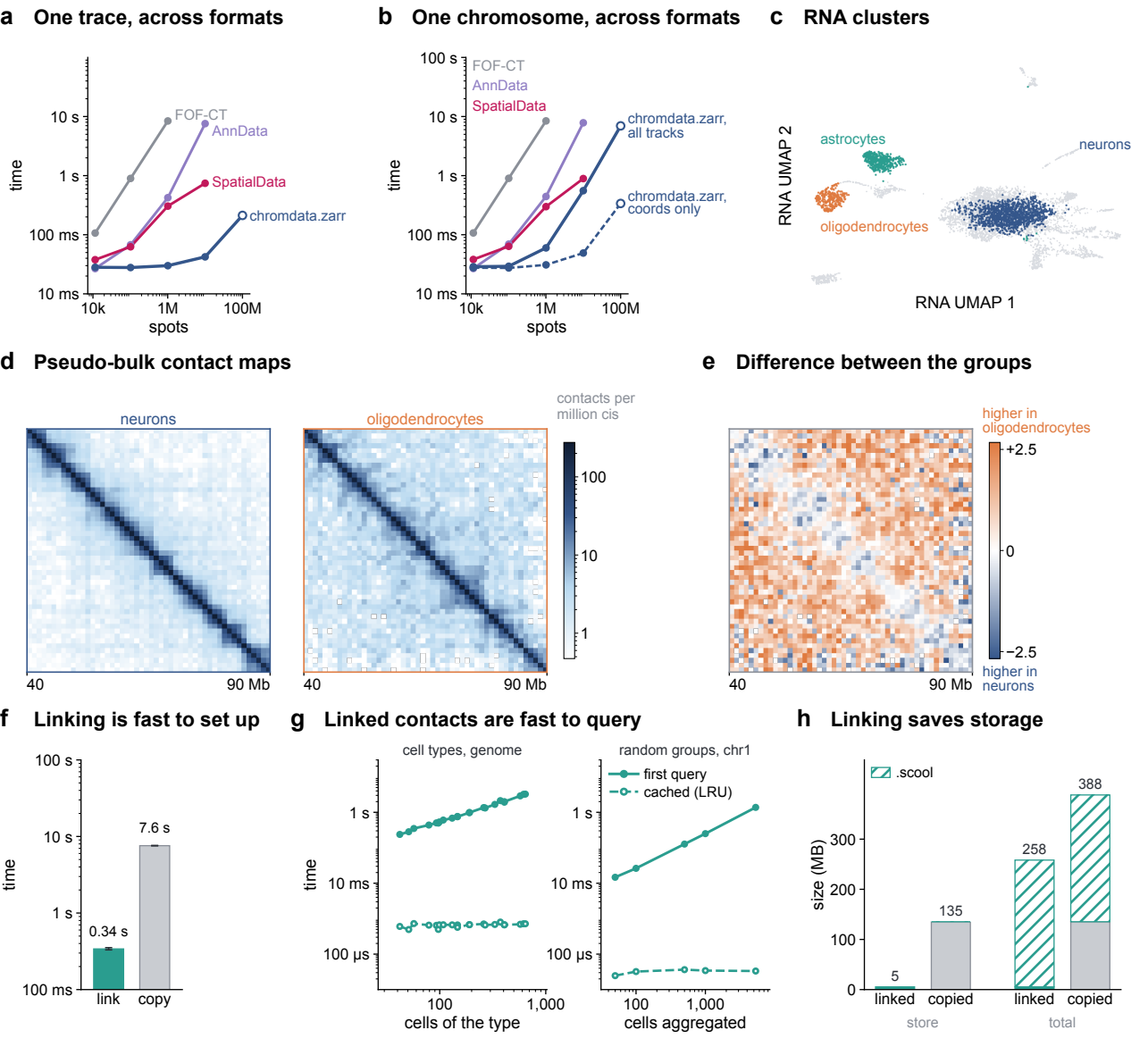

a One trace, across formats
time
10 s
1 s
100 ms
10 ms
FOF-CT
AnnData
SpatialData
chromdata.zarr
10k
1M
100M
spots
b One chromosome, across formats
time
100 s
10 s
1 s
100 ms
10 ms
FOF-CT
AnnData
SpatialData
chromdata.zarr, all tracks
chromdata.zarr, coords only
10k
1M
100M
spots
c RNA clusters
RNA UMAP 2
astrocytes
neurons
oligodendrocytes
RNA UMAP 1
d Pseudo-bulk contact maps
neurons
oligodendrocytes
contacts per million cis
100
10
1
40
90 Mb
40
90 Mb
e Difference between the groups
higher in oligodendrocytes
+2.5
0
−2.5
higher in neurons
40
90 Mb
f Linking is fast to set up
time
100 s
10 s
1 s
100 ms
7.6 s
0.34 s
link
copy
g Linked contacts are fast to query
cell types, genome
random groups, chr1
first query
cached (LRU)
time
1 s
10 ms
100 µs
100
1,000
cells of the type
cells aggregated
h Linking saves storage
.scool
size (MB)
300
200
100
0
388
258
135
5
linked
copied
linked
copied
store
total